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compact1, compact2, compact3

java.text

Class StringCharacterIterator

java.lang.Object
    java.text.StringCharacterIterator

All Implemented Interfaces:

`Cloneable`, `CharacterIterator`

```
public final class StringCharacterIterator
extends Object
implements CharacterIterator
```

`StringCharacterIterator` implements the `CharacterIterator` protocol for a `String`. The `StringCharacterIterator` class iterates over the entire `String`.

See Also:

`CharacterIterator`

Field Summary

Fields inherited from interface java.text.CharacterIterator

`DONE`

Constructor Summary

Constructors

Constructor	Description
`StringCharacterIterator(String text)`	

	Constructs an iterator with an initial index of 0.
`StringCharacterIterator(String text, int pos)`	Constructs an iterator with the specified initial index.
`StringCharacterIterator(String text, int begin, int end, int pos)`	Constructs an iterator over the given range of the given string, with the index set at the specified position.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`Object`	`clone()`	Creates a copy of this iterator.
`char`	`current()`	Implements CharacterIterator.current() for String.
`boolean`	`equals(Object obj)`	Compares the equality of two StringCharacterIterator objects.
`char`	`first()`	Implements CharacterIterator.first() for String.
`int`	`getBeginIndex()`	Implements CharacterIterator.getBeginIndex() for String.

int	**getEndIndex**()	Implements CharacterIterator.getEndIndex() for String.
int	**getIndex**()	Implements CharacterIterator.getIndex() for String.
int	**hashCode**()	Computes a hashcode for this iterator.
char	**last**()	Implements CharacterIterator.last() for String.
char	**next**()	Implements CharacterIterator.next() for String.
char	**previous**()	Implements CharacterIterator.previous() for String.
char	**setIndex**(int p)	Implements CharacterIterator.setIndex() for String.
void	**setText**(**String** text)	Reset this iterator to point to a new string.

Methods inherited from class java.lang.Object

finalize, getClass, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

StringCharacterIterator

```
public StringCharacterIterator(String text)
```

Constructs an iterator with an initial index of 0.

Parameters:

`text - the String to be iterated over`

StringCharacterIterator

```
public StringCharacterIterator(String text,
                               int pos)
```

Constructs an iterator with the specified initial index.

Parameters:

`text - The String to be iterated over`

`pos - Initial iterator position`

StringCharacterIterator

```
public StringCharacterIterator(String text,
                               int begin,
                               int end,
                               int pos)
```

Constructs an iterator over the given range of the given string, with the index set at the specified position.

Parameters:

`text - The String to be iterated over`

`begin - Index of the first character`

`end - Index of the character following the last character`

`pos - Initial iterator position`

Method Detail

setText

```
public void setText(String text)
```

Reset this iterator to point to a new string. This package-visible method is used by other java.text classes that want to avoid allocating new StringCharacterIterator objects every time their setText method is called.

Parameters:

`text - The String to be iterated over`

Since:

`1.2`

first

```
public char first()
```

Implements CharacterIterator.first() for String.

Specified by:

`first` in interface `CharacterIterator`

Returns:

the first character in the text, or DONE if the text is empty

See Also:

`CharacterIterator.first()`

last

```
public char last()
```

Implements CharacterIterator.last() for String.

Specified by:

`last` in interface `CharacterIterator`

Returns:

the last character in the text, or DONE if the text is empty

See Also:

`CharacterIterator.last()`

setIndex

```
public char setIndex(int p)
```

Implements CharacterIterator.setIndex() for String.

Specified by:

`setIndex` in interface `CharacterIterator`

Parameters:

`p - the position within the text. Valid values range from getBeginIndex() to getEndIndex(). An IllegalArgumentException is thrown if an invalid value is supplied.`

Returns:

`the character at the specified position or DONE if the specified position is equal to getEndIndex()`

See Also:

`CharacterIterator.setIndex(int)`

current

```
public char current()
```

Implements CharacterIterator.current() for String.

Specified by:

`current` in interface `CharacterIterator`

Returns:

`the character at the current position or DONE if the current position is off the end of the text.`

See Also:

`CharacterIterator.current()`

next

```
public char next()
```

Implements CharacterIterator.next() for String.

Specified by:

`next` in interface `CharacterIterator`

Returns:

`the character at the new position or DONE if the new position is off the end of the text range.`

See Also:

`CharacterIterator.next()`

previous

```
public char previous()
```

Implements CharacterIterator.previous() for String.

Specified by:

`previous` in interface `CharacterIterator`

Returns:

the character at the new position or DONE if the current position is equal to getBeginIndex().

See Also:

`CharacterIterator.previous()`

getBeginIndex

```
public int getBeginIndex()
```

Implements CharacterIterator.getBeginIndex() for String.

Specified by:

`getBeginIndex` in interface `CharacterIterator`

Returns:

the index at which the text begins.

See Also:

`CharacterIterator.getBeginIndex()`

getEndIndex

```
public int getEndIndex()
```

Implements CharacterIterator.getEndIndex() for String.

Specified by:

`getEndIndex` in interface `CharacterIterator`

Returns:

the index after the last character in the text

See Also:

`CharacterIterator.getEndIndex()`

getIndex

```
public int getIndex()
```

Implements CharacterIterator.getIndex() for String.

Specified by:

`getIndex` in interface `CharacterIterator`

Returns:

the current index.

See Also:

`CharacterIterator.getIndex()`

equals

```
public boolean equals(Object obj)
```

Compares the equality of two StringCharacterIterator objects.

Overrides:

`equals` in class `Object`

Parameters:

obj - the StringCharacterIterator object to be compared with.

Returns:

true if the given obj is the same as this StringCharacterIterator object; false otherwise.

See Also:

`Object.hashCode()`, `HashMap`

hashCode

```
public int hashCode()
```

Computes a hashcode for this iterator.

Overrides:

`hashCode` in class `Object`

Returns:

A hash code

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(java.lang.Object)`

clone

```
public Object clone()
```

Creates a copy of this iterator.

Specified by:

`clone` in interface `CharacterIterator`

Overrides:

`clone` in class `Object`

Returns:

A copy of this

See Also:

`Cloneable`

Submit a bug or feature

For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.

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